

Computational analysis of high carbon steel for optimal design

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Abstract—To increase the reliability of materials the computational analysis has been considered a vital tool for designing the components in the mechanical and nuclear engineering. By increasing a high percentage of carbon in liquid steel subsequently enhance the hardness, elasticity and toughness of the materials. In this research high carbon steel has been used due to its wear resistance properties in order to increasing hardness and strength. The findings of this research reveal the high carbon steel a good replacement material for conventional iron due to its low weight. Modeling and analysis of the steel, metallic glass and high carbon steel have been done using ANSYS software. Validation of the design has been done by applying a single vertical loads acting on the sheet, rectangular rod and cylindrical tube of high carbon steel having thickness 30, 60 and 90 mm. It is clear from the results that the overall weight of the structure has been reduced and improve load carrying capacity by using high carbon steel. This research highlights the replacement of the multi welded iron structure and the design constraints with the cylindrical tubes for bearing maximum stress having limiting displacement.

Keywords— *Hardness; Toughness; Deformation; Composite materials; Young's modulus; Percentage weight reduction; High carbon steel.*

I. INTRODUCTION

Due to load carrying capabilities, high temperature resistance, gradient microstructure, wear resistance and astonishing reliability the high carbon steel has been considered a best replacement of conventional iron. Considering these properties several studies concluded that the high carbon steel is one of the suitable manufacturing material for the industrial components, vehicle parts, aircraft, wind turbine blades and construction industry [1, 2]. Development of effective protection for vehicles requires a good understanding of impact phenomena and these are investigated by three main approaches;

- Analytical
- Experimental

- Numerical

All three procedures have its own merits and demerits. Experiments are expensive and cannot cover all the conditions that can occur in specified range. Analytical methods assume a lot of simplifications and only sufficient for a limited range of cases. Numerical results are limited to a single problem since the whole numerical procedure has to be repeated for any change in input variables [3].

Despite a number of studies had been carried out since last few decades to understand the properties of high carbon steel but still there is a need to explore more about the structural behavior of this material. Moreover, the mass, hardness and shapes were also considered [4] during experiments and simulation analysis to understand the response of high carbon steel in view of these effects. In the present study the high carbon steel has been tested in three different shapes (bar, sheet and cylinder) of different thickness (30, 60 and 90 mm).

In the past instead of high carbon steel the aluminum alloy had been tested by the bombardment of needle shaped objects but it was found that it breach the target. It was also noticed that the weight of the material has a great influence on the loading parameters and hence extensive material had been tested. Among these materials high carbon steel considered as one of the best candidate due to its high hardness, toughness and resistance against high temperature [5, 6].

The impact of load on different materials was investigated [7, 8] and it was observed that load depends on the properties of both incident and target material. The impact of load on rubber coating steel, low carbon steel and polymer based materials were also studied but in these cases the cracks appeared in the material and failures happened that affects the qualitative and quantitative analysis [9-14]. Keeping in view the same concept of load and strength another material named advanced high strength steel got the attention but this material did not showed sustainability against the high strength. Fractures and deformation were found during pressing and super tension process [15].